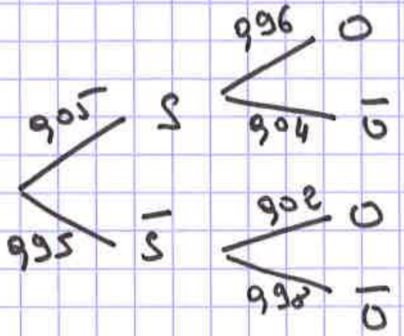
$p(S)=0{,}05$ $\qquad p(\bar S)=0{,}95$

$p_{\bar S}(O)=p(O|\bar S)=\frac{2}{100}=0{,}02$ (" faux positifs)....

$p_S(O)=p(O/_S)=0{,}96$



erreur : $S\cap\bar O$ ou $\bar S\cap O$

$p=0{,}05\times0{,}04+0{,}95\times0{,}02=0{,}003$. ($3/1000$).

Exercice 4

$T \approx U[0;1]$

1) $p(T>0{,}5)=\frac{1-0{,}5}{1-0}=0{,}5$

2) $p(0{,}2<T<0{,}6)=\frac{0{,}6-0{,}2}{1-0}=0{,}4$

3) $p(T>0{,}6)=\frac{1-0{,}6}{1-0}=0{,}4$

Ex 5

$p(X\leq 155)=0{,}0037$

↳ 0,4 %

$p(155\leq X\leq 175)=0{,}8103$

↳ 81,03 %

$X\in[165{,}28;\ 174{,}71]$

Exercice 7

```
def proba(a,b,x):
    return ((x-a)/(b-a))
```

Exercice 8

(114) $p(X>10)=\frac{20-10}{20-(-10)}=\frac{10}{30}=\frac{1}{3}$ Réponse (A) et (C)

(115) $E(X)=\frac{a+b}{2}=\frac{20+(-10)}{2}=5$ Réponse (B) et (C)

$f(x)=\frac{1}{20-(-10)}=\frac{1}{30}$ donc $E(X)=\int_{-10}^{20} x f(x)\,dx=\int_{-10}^{20}\frac{x}{30}\,dx$